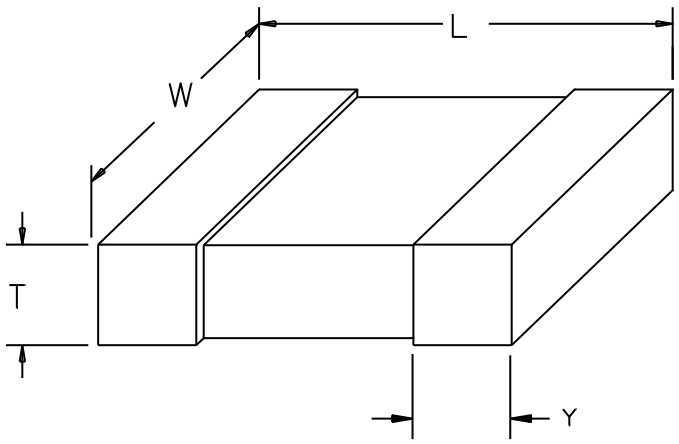
INCH-POUND
MIL-PRF-32535/3
w/AMENDMENT 1
31 July 2017
SUPERSEDING
MIL-PRF-32535/3
28 September 2015

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, CHIP, FIXED, CERAMIC DIELECTRIC (TEMPERATURE STABLE AND GENERAL PURPOSE), EXTENDED RANGE, HIGH RELIABILITY AND STANDARD RELIABILITY, SIZE 0603

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32535.




| Dimensions | | | |
|---|---|---|---|
| L ± .006 | W ± .006 | T Max. | Y ± .008 |
| .063 | .032 | .039 | .016 |

| inches | mm |
|---|---|
| .006 | 0.15 |
| .008 | 0.20 |
| .016 | 0.41 |
| .032 | 0.81 |
| .039 | 0.99 |
| .063 | 1.60 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimensions and tolerances are for terminated chips.
4. For solder termination finishes, add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerances. The increase in dimension applies to the solder coating thickness only.

FIGURE 1. Size 0603 capacitors.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Voltage-temperature limit or temperature characteristic (VTL/TC): BP, C0G, and X7R as specified in MIL-PRF-32535.

Rated voltage ($V_{dc}$): V = 4; W = 6.3; X = 10; Y = 16; Z = 25; A = 50; B = 100; C = 200. See table I for maximum rated voltage available for each capacitance value.

Operating temperature range: -55°C to +125°C.

Termination finish: D, G, M, R, V, and Z as specified in MIL-PRF-32535.

Electrode: P and B as specified in MIL-PRF-32535.

Product level designator: Standard reliability – M and high reliability - T.

Marking: Package marking only in accordance with MIL-PRF-32535.

TABLE I. Size 0603 capacitor characteristics.

| Part or Identifying Number (PIN) 1/ | Capacitance (pF) | Capacitance tolerance | VTL/TC | Rated voltage 2/ ($V_{dc}$) | Electrode material |
|---|---|---|---|---|---|
| M3253503 - - - 1R0 - - - - | 1.0 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 1R5 - - - - | 1.5 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 2R2 - - - - | 2.2 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 2R7 - - - - | 2.7 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 3R3 - - - - | 3.3 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 3R9 - - - - | 3.9 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 4R7 - - - - | 4.7 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 5R6 - - - - | 5.6 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 6R8 - - - - | 6.8 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 8R2 - - - - | 8.2 | B, C, D | BP, C0G | 200 | P, B |
| M3253503 - - - 100 - - - - | 10 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 120 - - - - | 12 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 150 - - - - | 15 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 180 - - - - | 18 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 200 - - - - | 20 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 220 - - - - | 22 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 270 - - - - | 27 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 330 - - - - | 33 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 390 - - - - | 39 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 470 - - - - | 47 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 560 - - - - | 56 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 680 - - - - | 68 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 820 - - - - | 82 | F, G, J, K | BP, C0G | 200 | P, B |
| M3253503 - - - 101 - - - - | 100 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 121 - - - - | 120 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 151 - - - - | 150 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 181 - - - - | 180 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 221 - - - - | 220 | F, G, J, K | BP, C0G | 100 | P, B |

See footnotes at end of table.



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TABLE I. Size 0603 capacitor characteristics - Continued.

| Part or Identifying Number (PIN) 1/ | Capacitance (pF) | Capacitance tolerance | VTL/TC | Rated voltage 2/ ($V_{dc}$) | Electrode material |
|---|---|---|---|---|---|
| M3253503 - - - 271 - - - - | 270 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 331 - - - - | 330 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 391 - - - - | 390 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 471 - - - - | 470 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 561 - - - - | 560 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 681 - - - - | 680 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 821 - - - - | 820 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 102 - - - - | 1,000 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 122 - - - - | 1,200 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 152 - - - - | 1,500 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 182 - - - - | 1,800 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 202 - - - - | 2,000 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 222 - - - - | 2,200 | F, G, J, K | BP, C0G | 100 | P, B |
| M3253503 - - - 272 - - - - | 2,700 | F, G, J, K | BP, C0G | 50 | P, B |
| M3253503 - - - 332 - - - - | 3,300 | F, G, J, K | BP, C0G | 50 | P, B |
| M3253503 - - - 392 - - - - | 3,900 | F, G, J, K | BP, C0G | 50 | P, B |
| M3253503 - - - 472 - - - - | 4,700 | F, G, J, K | BP, C0G | 25 | P, B |
| M3253503 - - - 562 - - - - | 5,600 | F, G, J, K | BP, C0G | 25 | P, B |
| M3253503E2 - 102 - - - - | 1,000 | K, M | X7R | 100 | P, B |
| M3253503E2 - 122 - - - - | 1,200 | K, M | X7R | 100 | P, B |
| M3253503E2 - 152 - - - - | 1,500 | K, M | X7R | 100 | P, B |
| M3253503E2 - 182 - - - - | 1,800 | K, M | X7R | 100 | P, B |
| M3253503E2 - 222 - - - - | 2,200 | K, M | X7R | 100 | P, B |
| M3253503E2 - 272 - - - - | 2,700 | K, M | X7R | 100 | P, B |
| M3253503E2 - 332 - - - - | 3,300 | K, M | X7R | 100 | P, B |
| M3253503E2 - 392 - - - - | 3,900 | K, M | X7R | 100 | P, B |
| M3253503E2 - 472 - - - - | 4,700 | K, M | X7R | 100 | P, B |
| M3253503E2 - 562 - - - - | 5,600 | K, M | X7R | 100 | P, B |
| M3253503E2 - 682 - - - - | 6,800 | K, M | X7R | 100 | P, B |
| M3253503E2 - 822 - - - - | 8,200 | K, M | X7R | 100 | P, B |
| M3253503E2 - 103 - - - - | 10,000 | K, M | X7R | 100 | P, B |
| M3253503E2 - 123 - - - - | 12,000 | K, M | X7R | 100 | P, B |
| M3253503E2 - 153 - - - - | 15,000 | K, M | X7R | 100 | P, B |
| M3253503E2 - 183 - - - - | 18,000 | K, M | X7R | 100 | P, B |
| M3253503E2 - 223 - - - - | 22,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 273 - - - - | 27,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 333 - - - - | 33,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 393 - - - - | 39,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 473 - - - - | 47,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 513 - - - - | 51,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 563 - - - - | 56,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 683 - - - - | 68,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 823 - - - - | 82,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 104 - - - - | 100,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 124 - - - - | 120,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 154 - - - - | 150,000 | K, M | X7R | 50 | P, B |
| M3253503E2 - 184 - - - - | 180,000 | K, M | X7R | 25 | P, B |
| M3253503E2 - 224 - - - - | 220,000 | K, M | X7R | 16 | P, B |

1/ The complete PIN shall include additional symbols to indicate VTL/TC (where applicable), voltage, capacitance tolerance, termination finish, product level, and electrode material.
2/ This is the maximum rated voltage available. All lower voltage ratings are also available.



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Navy - EC
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Preparing activity:
DLA - CC

(Project 5910-2017-033)

Review activities:
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Navy - AS, MC, OS, SH
Air Force - 19, 99
Other – MDA, NA

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